Week 5 – Splunk Detection Search Queries

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SEARCH 01: BRUTE FORCE DETECTION

1. Brute Force Detection

MITRE ATT&CK: Potential mapping to T1110

Detection Objective: Identify multiple authentication failures grouped by host, process, and user account to detect brute-force password guessing attempts.

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") | stats count as failed_attempts by host, process, logname | sort - failed_attempts
```

What the Query Does:

- `index=main sourcetype=linux_secure` — Restricts search to Linux authentication logs in the main index.
- `("authentication failure" OR "Failed password")` — Filters raw log strings for failed logon attempts.
- `stats count as failed_attempts by host, process, logname` — Aggregates failure counts per host, binary, and username.
- `sort - failed_attempts` — Orders output descending by highest failure count.

Actual Search Result: 4 events found (2 aggregated rows for user `gunjan`: `gdm-password` = 2 failures, `sudo` = 1 failure).

SOC Analyst Interpretation: Failures originated from local processes (`gdm-password`, `sudo`) for user *gunjan* on *ubuntu-soc* without remote IPs, indicating local user typo rather than an external brute-force attack.

Limitations: Lacks remote network IP address fields in the current statistical aggregation.

SEARCH 02: NEW ACCOUNT CREATION DETECTION

2. New Account Creation Detection

MITRE ATT&CK: T1136 — Create Account

Detection Objective: Detect newly created local user accounts on Linux endpoints to identify unauthorized persistence or backdoor account creation.

```
index=main sourcetype=linux_secure ("useradd" OR "adduser") | table _time host _raw | sort -_time
```

What the Query Does:

- `index=main sourcetype=linux_secure` — Queries Linux security logs.
- `("useradd" OR "adduser")` — Searches for execution keywords associated with Linux account creation.
- `table _time host _raw` — Displays event timestamp, hostname, and full raw log string.
- `sort -_time` — Displays most recent account creation events at top.

Actual Search Result: 1 event found (Timestamp: `2026-11-05 12:40:11.545`, Host: `ubuntu-soc`, Raw: `new user: name=splunk, UID=1001, GID=1001`).

SOC Analyst Interpretation: Confirms a new user account named `splunk` was created. Aligns with normal administrator setup during lab deployment.

Limitations: Query does not extract initiating user ID into a dedicated field without parsing `_raw`.

SEARCH 03: SUSPICIOUS POWERSHELL ACTIVITY

3. Suspicious PowerShell Activity

MITRE ATT&CK: T1059.001 — PowerShell

Detection Objective: Identify execution of Windows PowerShell instances and command-line parameters indicative of malicious script execution.

```
index=main sourcetype=linux_secure ("useradd" OR "adduser") | search Image="*powershell.exe*" | tab
le _time host user Image CommandLine | sort -_time
```

What the Query Does:

- `index=main sourcetype=linux_secure` — Targets dataset log stream.
- `search Image="*powershell.exe*"` — Filters events for process execution matching PowerShell executable names.
- `table _time host user Image CommandLine` — Formats output with execution time, username, image path, and arguments.

Actual Search Result: No matching events found in the available dataset (0 events).

SOC Analyst Interpretation: Zero results reflect that the environment is a Linux host (`ubuntu-soc`) where Windows Sysmon/Event ID 1 process creation logs are not ingested. Does not prove PowerShell execution did not occur elsewhere.

Limitations: Windows Sysmon / Event ID 4688 telemetry is not ingested in this Linux log pipeline.

SEARCH 04: NETWORK SERVICE SCANNING DETECTION

4. Network Service Scanning Detection

MITRE ATT&CK: T1046 — Network Service Discovery

Detection Objective: Identify network port scanning activity, SYN scans, or Nmap execution across indexed log sources.

```
index=* ("SYN Scan" OR "port scan" OR "Nmap") | table _time host source src_ip dest_ip _raw | sort
-_time
```

What the Query Does:

- `index=*` — Searches across all accessible indexes.
- `("SYN Scan" OR "port scan" OR "Nmap")` — Filters for strings generated by network IDSs or firewalls.
- `table _time host source src_ip dest_ip _raw` — Formats output with source/destination IPs and raw messages.

Actual Search Result: No matching events found in the available dataset (0 events).

SOC Analyst Interpretation: No network scan alerts detected because firewall and Suricata/Snort IDS traffic logs are not currently forwarded to Splunk. Zero results cannot rule out unmonitored network scanning.

Limitations: Lack of network IDS / firewall telemetry integration in the lab SIEM.

SEARCH 05: PHISHING ACTIVITY DETECTION

5. Phishing Activity Detection

MITRE ATT&CK: T1566 — Phishing

Detection Objective: Detect email-borne threats, suspicious attachment keywords, or phishing indicators in indexed data.

```
index=* ("phishing" OR "phish" OR "suspicious attachment") | table _time host source _raw | sort -_
time
```

What the Query Does:

- `index=*` — Scans all available indexes.
- `("phishing" OR "phish" OR "suspicious attachment")` — Filters for string indicators associated with email gateways.
- `table _time host source _raw` — Outputs relevant timestamp, source log file, and raw message text.

Actual Search Result: No matching events found in the available dataset (0 events).

SOC Analyst Interpretation: No phishing logs retrieved. Mail server (Exchange/Postfix) and Secure Email Gateway (SEG) logs are not present in this endpoint-centric setup.

Limitations: Absence of email gateway and mail server log ingestion.

SEARCH 06: LINUX AUTHENTICATION FAILURE ANALYSIS

6. Linux Authentication Failure Analysis

MITRE ATT&CK: Potential mapping to T1110 (Evaluated with Search 01)

Detection Objective: Provide detailed user and process distribution analysis for authentication failures to isolate specific failing services.

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") | stats count as
failure_count values(process) as processes by host, logname | sort - failure_count
```

What the Query Does:

- `stats count as failure_count` — Calculates total failure count per group.
- `values(process) as processes` — Displays a distinct list of processes associated with failures.
- `by host, logname` — Groups failure counts by hostname and targeted Linux account name.

Actual Search Result: 4 events found (1 aggregated row for user `gunjan` on host `ubuntu-soc` with `failure_count` = 3 and `processes` = `gdm-password, sudo`).

SOC Analyst Interpretation: Correlates with Search 01, confirming user *gunjan* experienced 3 authentication failures spanning display manager logon and privilege escalation. Represents normal local user operation.

Limitations: Evaluates same endpoint scope as Search 01; does not capture PAM module parameters.

SEARCH 07: PRIVILEGED COMMAND MONITORING

7. Privileged Command Monitoring

MITRE ATT&CK: Contextual Audit (No fixed mapping)

Detection Objective: Monitor execution of administrative commands using `sudo` or `pkexec` to audit privileged user activity.

```
index=main sourcetype=linux_secure "COMMAND=" | table _time host _raw | sort -_time
```

What the Query Does:

- `"COMMAND="` — Filters Linux security logs for audit strings recorded when privileged commands execute.
- `table _time host _raw` — Formats output chronologically with timestamps and raw execution details.

Actual Search Result: 33 events found (Including `pkexec` package notifier checks and `sudo splunk start/status` commands by user `gunjan`).

SOC Analyst Interpretation: Demonstrates active system administration. Executed commands (such as managing Splunk services and system updates) reflect standard administrative tasks without malicious indicators.

Limitations: Detailed command arguments require manual parsing of raw log text strings.